

COMP5461 – OS

Programming assignment 2 – Results and Observations

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Phase 0:**Unsynchronized run:**

Activating the network ...

Initializing the server ...

Initializing the Accounts database ...

Connecting server to network ...

Initializing client sending application ...

Initializing the transactions ...

Connecting client to network ...

Initializing client receiving application ...

DEBUG : Client.run() - starting client receiving thread connected

DEBUG : Client.run() - starting client sending thread connected

Server 2 starting

Server 1 starting

DEBUG : Server.withdraw - i 1 Current balance 1000.0 Amount 500.0 1

DEBUG : Server.deposit - i 0 Current balance 10.0 Amount 90.0 2

DEBUG : Server.withdraw - i 2 Current balance 28000.0 Amount 20000.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 11528

DEBUG : Server.deposit - i 3 Current balance 120000.0 Amount 20000.0 2

DEBUG : Server.withdraw - i 4 Current balance 125.0 Amount 25.0 1

DEBUG : Server.deposit - i 6 Current balance 17000.0 Amount 3000.0 1

Account number 11528 Account Balance 500.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31718

Account number 31718 Account Balance 100.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 16783

Account number 16783 Account Balance 8000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 10538

Account number 10538 Account Balance 140000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21015

Account number 21015 Account Balance 100.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 82465

DEBUG : Server.query - i 7 Current balance 20500.0 1

DEBUG : Server.query - i 5 Current balance 1750.0 2

Account number 82465 Account Balance 20000.0 Message done

DEBUG : Server.deposit - i 8 Current balance 75.0 Amount 25.0 1

DEBUG : Server.withdraw - i 9 Current balance 1300.0 Amount 1500.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31816

DEBUG : Server.deposit - i 11 Current balance 25.0 Amount 725.0 1

DEBUG : Server.query - i 12 Current balance 1000.0 2

Account number 31816 Account Balance 20500.0 Message done

DEBUG : Server.deposit - i 13 Current balance 1000.0 Amount 100.0 1

DEBUG : Server.withdraw - i 14 Current balance 100.0 Amount 100.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61326

DEBUG : Server.deposit - i 15 Current balance 45.0 Amount 55.0 1

Account number 61326 Account Balance 1750.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 73745

Account number 73745 Account Balance 100.0 Message done

DEBUG : Server.withdraw - i 16 Current balance 0.0 Amount 1000.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 81514

DEBUG : Server.withdraw - i 17 Current balance 38000.0 Amount 5000.0 1

Account number 81514 Account Balance -200.0 Message done

DEBUG : Server.query - i 18 Current balance 65000.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92012

Account number 92012 Account Balance 750.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 52853

Account number 52853 Account Balance 1000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 41714

Account number 41714 Account Balance 1100.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 58342

Account number 58342 Account Balance 0.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 62080

Account number 62080 Account Balance 100.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21638

Account number 21638 Account Balance -1000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 52726

Account number 52726 Account Balance 33000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91316

Account number 91316 Account Balance 65000.0 Message done

DEBUG : Server.deposit - i 19 Current balance 65.0 Amount 1035.0 1

DEBUG : Server.deposit - i 9 Current balance -200.0 Amount 200.0 2

DEBUG : Server.deposit - i 21 Current balance 1280.0 Amount 220.0 1

DEBUG : Server.withdraw - i 22 Current balance 10.0 Amount 10.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 60520

DEBUG : Server.withdraw - i 23 Current balance 2850.0 Amount 1850.0 1

DEBUG : Server.deposit - i 24 Current balance 50.0 Amount 250.0 2

Account number 60520 Account Balance 1100.0 Message done

DEBUG : Server.withdraw - i 19 Current balance 1100.0 Amount 100.0 2

DEBUG : Server.withdraw - i 25 Current balance 0.0 Amount 800.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 81514

DEBUG : Server.withdraw - i 26 Current balance 1100.0 Amount 100.0 2

DEBUG : Server.deposit - i 27 Current balance 5200.0 Amount 1800.0 1

Account number 81514 Account Balance 0.0 Message done

DEBUG : Server.query - i 28 Current balance 2000.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 41911

DEBUG : Server.deposit - i 31 Current balance 1005.0 Amount 495.0 2

Account number 41911 Account Balance 1500.0 Message done

DEBUG : Server.deposit - i 32 Current balance 5300.0 Amount 500.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 39331

DEBUG : Server.query - i 33 Current balance 0.0 2

Account number 39331 Account Balance 0.0 Message done

DEBUG : Server.withdraw - i 34 Current balance 1090.0 Amount 90.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91312

DEBUG : Server.withdraw - i 35 Current balance 100.0 Amount 500.0 2

Account number 91312 Account Balance 1000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 15307

DEBUG : Server.withdraw - i 29 Current balance 20.0 Amount 50.0 2

Account number 15307 Account Balance 300.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 60520

DEBUG : Server.deposit - i 36 Current balance 85.0 Amount 15.0 2

Account number 60520 Account Balance 1000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 51728

DEBUG : Server.deposit - i 37 Current balance 675.0 Amount 325.0 2

Account number 51728 Account Balance -800.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 43902

DEBUG : Server.withdraw - i 38 Current balance 1900.0 Amount 900.0 2

Account number 43902 Account Balance 1000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 52521

Account number 52521 Account Balance 7000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31215

Account number 31215 Account Balance 2000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 41024

Account number 41024 Account Balance 1500.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 38421

Account number 38421 Account Balance 5800.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 70321

Account number 70321 Account Balance 0.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 12421

Account number 12421 Account Balance 1000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 82411

Account number 82411 Account Balance -400.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 22310

Account number 22310 Account Balance -30.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21117

Account number 21117 Account Balance 100.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 00925

Account number 00925 Account Balance 1000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31820

Account number 31820 Account Balance 1000.0 Message done

DEBUG : Server.deposit - i 29 Current balance 20.0 Amount 80.0 1

DEBUG : Server.deposit - i 39 Current balance 65.0 Amount 35.0 2

DEBUG : Server.deposit - i 41 Current balance 1500.0 Amount 1500.0 1

DEBUG : Server.withdraw - i 42 Current balance 200.0 Amount 300.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 22310

DEBUG : Server.deposit - i 43 Current balance 250.0 Amount 250.0 1

DEBUG : Server.deposit - i 44 Current balance 95.0 Amount 305.0 2

Account number 22310 Account Balance 100.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91715

DEBUG : Server.deposit - i 45 Current balance 0.0 Amount 1000.0 1

DEBUG : Server.withdraw - i 39 Current balance 100.0 Amount 150.0 2

Account number 91715 Account Balance 100.0 Message done

DEBUG : Server.withdraw - i 47 Current balance 10000.0 Amount 10000.0 2

DEBUG : Server.deposit - i 48 Current balance 35.0 Amount 65.0 2

DEBUG : Server.query - i 46 Current balance 1620.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 32613

DEBUG : Server.withdraw - i 49 Current balance 0.0 Amount 250.0 2

DEBUG : Server.withdraw - i 51 Current balance 1345.0 Amount 345.0 1

Account number 32613 Account Balance 3000.0 Message done

DEBUG : Server.deposit - i 52 Current balance 145.0 Amount 155.0 2

DEBUG : Server.deposit - i 53 Current balance 300.0 Amount 700.0 1

DEBUG : Server.deposit - i 54 Current balance 560.0 Amount 300.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 13133

Account number 13133 Account Balance -100.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 43018

DEBUG : Server.withdraw - i 55 Current balance 1850.0 Amount 350.0 2

Account number 43018 Account Balance 500.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91412

Account number 91412 Account Balance 400.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 11314

Account number 11314 Account Balance 1000.0 Message done

DEBUG : Server.deposit - i 49 Current balance -250.0 Amount 350.0 2

DEBUG : Server.deposit - i 49 Current balance -250.0 Amount 350.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92018

DEBUG : Server.deposit - i 57 Current balance 1420.0 Amount 180.0 2

DEBUG : Server.withdraw - i 58 Current balance 3500.0 Amount 2300.0 1

Account number 92018 Account Balance 100.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92018

DEBUG : Server.withdraw - i 59 Current balance 980.0 Amount 900.0 2

DEBUG : Server.query - i 61 Current balance 1.05E7 1

DEBUG : Server.withdraw - i 62 Current balance 1740.0 Amount 1500.0 2

DEBUG : Server.query - i 63 Current balance 535000.0 1

DEBUG : Server.deposit - i 64 Current balance 845250.0 Amount 134750.0 2

DEBUG : Server.withdraw - i 65 Current balance 375.0 Amount 250.0 1

DEBUG : Server.withdraw - i 59 Current balance 80.0 Amount 80.0 2

Account number 92018 Account Balance 100.0 Message done

DEBUG : Server.deposit - i 67 Current balance 1290.0 Amount 1110.0 2

DEBUG : Server.deposit - i 66 Current balance 50000.0 Amount 50000.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91214

DEBUG : Server.withdraw - i 69 Current balance 625.0 Amount 600.0 1

DEBUG : Server.deposit - i 68 Current balance 350.0 Amount 650.0 2

Account number 91214 Account Balance 1600.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91245

Account number 91245 Account Balance 1200.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42018

Account number 42018 Account Balance 80.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 01314

Account number 01314 Account Balance 1.05E7 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 20922

Account number 20922 Account Balance 240.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21219

Account number 21219 Account Balance 535000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 83530

Account number 83530 Account Balance 980000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 83830

Account number 83830 Account Balance 125.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42018

Account number 42018 Account Balance 0.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92318

Account number 92318 Account Balance 2400.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 72820

Account number 72820 Account Balance 100000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42520

Account number 42520 Account Balance 1000.0 Message done

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61612

Account number 61612 Account Balance 25.0 Message done

Terminating client sending thread - Running time 362 milliseconds

DEBUG : Server.deposit - i 69 Current balance 25.0 Amount 75.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61612

Account number 61612 Account Balance 100.0 Message done

Process finished with exit code 130

Comments:

The program hangs because lack of mutual exclusion in 2 threads executing a section like the one highlighted below will end up corrupting the counter. The program will wait endlessly because the counter never reaches the target value. In other words, 2 transactions completing concurrently would cause the counter to decrement by 1 instead of 2 forcing the process to wait even if all transactions are completed.

```
while (Network.getOutBufferStatus().equals("full"))
{
    Thread.yield();    /* Yield the cpu if the network output buffer is full /
}

    / System.out.println("\n DEBUG : Server.processTransactions() - transferring out account "
+ trans.getAccountNumber()); /

    Network.transferOut(trans);                / Transfer a completed transaction from
the server to the network output buffer /
    setNumberOfTransactions( (getNumberOfTransactions() + 1) );    / Count the number of
transactions processed */
```

Phase 1:**Synchronized methods:**

Activating the network ...

Initializing the server ...

Initializing the Accounts database ...

Connecting server to network ...

Initializing client sending application ...

Initializing the transactions ...

Connecting client to network ...

Initializing client receiving application ...

DEBUG : Client.run() - starting client sending thread connected
Server 2 starting

DEBUG : Client.run() - starting client receiving thread connected
Server 1 starting

SERVER 2 received 31718

Exception in thread "Thread-2" SERVER 1 received 11528

java.lang.Error: Unresolved compilation problem:

Cannot make a static reference to the non-static method getServerThreadId() from the type Server

Receiving transaction 0/71

at Server.deposit(Server.java:391)

at Server.processTransactions(Server.java:278)

at Server.run(Server.java:514)

Exception in thread "Thread-1" java.lang.Error: Unresolved compilation problem:

Cannot make a static reference to the non-static method getServerThreadId() from the type Server

at Server.withdraw(Server.java:429)

at Server.processTransactions(Server.java:289)

at Server.run(Server.java:514)

Synchrhonized statements:

Activating the network ...

Initializing the server ...

Inializing the Accounts database ...

Connecting server to network ...

Initializing client sending application ...

Initializing the transactions ...

Connecting client to network ...

Initializing client receiving application ...

DEBUG : Client.run() - starting client receiving thread connected

Server 1 starting

Server 2 starting

DEBUG : Client.run() - starting client sending thread connected

Receiving transaction 0/71

SERVER 1 received 31718

SERVER 2 received 31718

DEBUG : Server.deposit - i 0 Current balance 10.0 Amount 90.0 1

DEBUG : Server.deposit - i 0 Current balance 100.0 Amount 90.0 2

DEBUG : Server.processTransactions() - transferring out account 31718

DEBUG : Server.processTransactions() - transferring out account 31718

SERVER 1 received 16783

SERVER 2 received 10538

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31718

DEBUG : Server.withdraw - i 2 Current balance 28000.0 Amount 20000.0 1

DEBUG : Server.processTransactions() - transferring out account 16783

DEBUG : Server.deposit - i 3 Current balance 120000.0 Amount 20000.0 2

DEBUG : Server.processTransactions() - transferring out account 10538

SERVER 1 received 21015

SERVER 2 received 61326

DEBUG : Server.withdraw - i 4 Current balance 125.0 Amount 25.0 1

DEBUG : Server.processTransactions() - transferring out account 21O15

Account number 31718 Account Balance 100.0 Message done
Receiving transaction 1/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31718

Account number 31718 Account Balance 190.0 Message done
Receiving transaction 2/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 16783

Account number 16783 Account Balance 8000.0 Message done
Receiving transaction 3/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 10538

Account number 10538 Account Balance 140000.0 Message done
Receiving transaction 4/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21O15

Account number 21O15 Account Balance 100.0 Message done
Receiving transaction 5/71

DEBUG : Server.query - i 5 Current balance 1750.0 2

DEBUG : Server.processTransactions() - transferring out account 61326
SERVER 1 received 82465

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61326

DEBUG : Server.deposit - i 6 Current balance 17000.0 Amount 3000.0 1
SERVER 2 received 31816

Account number 61326 Account Balance 1750.0 Message done

DEBUG : Server.processTransactions() - transferring out account 82465

DEBUG : Server.query - i 7 Current balance 20500.0 2
Receiving transaction 6/71
SERVER 1 received 73745

DEBUG : Server.processTransactions() - transferring out account 31816

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 82465

DEBUG : Server.deposit - i 8 Current balance 75.0 Amount 25.0 1

Account number 82465 Account Balance 20000.0 Message done

DEBUG : Server.processTransactions() - transferring out account 73745
SERVER 2 received 81514
Receiving transaction 7/71
SERVER 1 received 92012

DEBUG : Server.withdraw - i 9 Current balance 1300.0 Amount 1500.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31816

DEBUG : Server.processTransactions() - transferring out account 81514

DEBUG : Server.deposit - i 11 Current balance 25.0 Amount 725.0 1

Account number 31816 Account Balance 20500.0 Message done

DEBUG : Server.processTransactions() - transferring out account 92012
SERVER 2 received 52853
Receiving transaction 8/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 73745

DEBUG : Server.query - i 12 Current balance 1000.0 2
SERVER 1 received 41714

Account number 73745 Account Balance 100.0 Message done

DEBUG : Server.processTransactions() - transferring out account 52853
Receiving transaction 9/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 81514

DEBUG : Server.deposit - i 13 Current balance 1000.0 Amount 100.0 1

Account number 81514 Account Balance -200.0 Message done
SERVER 2 received 58342

DEBUG : Server.processTransactions() - transferring out account 41714
Receiving transaction 10/71

DEBUG : Server.withdraw - i 14 Current balance 100.0 Amount 100.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92012

DEBUG : Server.processTransactions() - transferring out account 58342
SERVER 1 received 62080

Account number 92012 Account Balance 750.0 Message done
SERVER 2 received 81514
Receiving transaction 11/71

DEBUG : Server.deposit - i 15 Current balance 45.0 Amount 55.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 52853

DEBUG : Server.processTransactions() - transferring out account 62080

Account number 52853 Account Balance 1000.0 Message done
Receiving transaction 12/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 41714

Account number 41714 Account Balance 1100.0 Message done
Receiving transaction 13/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 58342

Account number 58342 Account Balance 0.0 Message done
Receiving transaction 14/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 62080

Account number 62080 Account Balance 100.0 Message done
Receiving transaction 15/71

DEBUG : Server.deposit - i 9 Current balance -200.0 Amount 200.0 2

DEBUG : Server.processTransactions() - transferring out account 81514
SERVER 1 received 21638

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 81514
SERVER 2 received 52726

DEBUG : Server.withdraw - i 16 Current balance 0.0 Amount 1000.0 1

Account number 81514 Account Balance 0.0 Message done

DEBUG : Server.processTransactions() - transferring out account 21638

DEBUG : Server.withdraw - i 17 Current balance 38000.0 Amount 5000.0 2
Receiving transaction 16/71

DEBUG : Server.processTransactions() - transferring out account 52726
SERVER 1 received 91316
SERVER 2 received 60520

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21638

DEBUG : Server.query - i 18 Current balance 65000.0 1

Account number 21638 Account Balance -1000.0 Message done

DEBUG : Server.processTransactions() - transferring out account 91316
Receiving transaction 17/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 52726

Account number 52726 Account Balance 33000.0 Message done
Receiving transaction 18/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91316

Account number 91316 Account Balance 65000.0 Message done
Receiving transaction 19/71

DEBUG : Server.deposit - i 19 Current balance 65.0 Amount 1035.0 2

DEBUG : Server.processTransactions() - transferring out account 60520
SERVER 1 received 41911
SERVER 2 received 39331

DEBUG : Server.deposit - i 21 Current balance 1280.0 Amount 220.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 60520

DEBUG : Server.processTransactions() - transferring out account 41911

Account number 60520 Account Balance 1100.0 Message done

DEBUG : Server.withdraw - i 22 Current balance 10.0 Amount 10.0 2
Receiving transaction 20/71
SERVER 1 received 91312

DEBUG : Server.processTransactions() - transferring out account 39331

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 41911

DEBUG : Server.withdraw - i 23 Current balance 2850.0 Amount 1850.0 1

Account number 41911 Account Balance 1500.0 Message done
Receiving transaction 21/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 39331

DEBUG : Server.processTransactions() - transferring out account 91312

SERVER 2 received 15307

Account number 39331 Account Balance 0.0 Message done
Receiving transaction 22/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91312
SERVER 1 received 51728

Account number 91312 Account Balance 1000.0 Message done
Receiving transaction 23/71

DEBUG : Server.deposit - i 24 Current balance 50.0 Amount 250.0 2

DEBUG : Server.processTransactions() - transferring out account 15307

DEBUG : Server.withdraw - i 25 Current balance 0.0 Amount 800.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 15307

DEBUG : Server.processTransactions() - transferring out account 51728
SERVER 2 received 60520

Account number 15307 Account Balance 300.0 Message done
Receiving transaction 24/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 51728
SERVER 1 received 43902

Account number 51728 Account Balance -800.0 Message done
Receiving transaction 25/71

DEBUG : Server.withdraw - i 19 Current balance 1100.0 Amount 100.0 2

DEBUG : Server.processTransactions() - transferring out account 60520

DEBUG : Server.withdraw - i 26 Current balance 1100.0 Amount 100.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 60520

DEBUG : Server.processTransactions() - transferring out account 43902
SERVER 2 received 52521

Account number 60520 Account Balance 1000.0 Message done
SERVER 1 received 31215
Receiving transaction 26/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 43902

DEBUG : Server.deposit - i 27 Current balance 5200.0 Amount 1800.0 2

Account number 43902 Account Balance 1000.0 Message done
Receiving transaction 27/71

DEBUG : Server.query - i 28 Current balance 2000.0 1

DEBUG : Server.processTransactions() - transferring out account 52521

DEBUG : Server.processTransactions() - transferring out account 31215

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 52521
SERVER 2 received 22310
SERVER 1 received 41024

Account number 52521 Account Balance 7000.0 Message done
Receiving transaction 28/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31215

Account number 31215 Account Balance 2000.0 Message done
Receiving transaction 29/71

DEBUG : Server.deposit - i 29 Current balance 20.0 Amount 80.0 2

DEBUG : Server.processTransactions() - transferring out account 22310

DEBUG : Server.deposit - i 31 Current balance 1005.0 Amount 495.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 22310

DEBUG : Server.processTransactions() - transferring out account 41024

Account number 22310 Account Balance 100.0 Message done
SERVER 1 received 70321
SERVER 2 received 38421
Receiving transaction 30/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 41024

DEBUG : Server.query - i 33 Current balance 0.0 1

Account number 41024 Account Balance 1500.0 Message done

DEBUG : Server.processTransactions() - transferring out account 70321
Receiving transaction 31/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 70321

Account number 70321 Account Balance 0.0 Message done

Receiving transaction 32/71

DEBUG : Server.deposit - i 32 Current balance 5300.0 Amount 500.0 2
SERVER 1 received 12421

DEBUG : Server.processTransactions() - transferring out account 38421

DEBUG : Server.withdraw - i 34 Current balance 1090.0 Amount 90.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 38421

DEBUG : Server.processTransactions() - transferring out account 12421

Account number 38421 Account Balance 5800.0 Message done
Receiving transaction 33/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 12421
SERVER 2 received 82411
SERVER 1 received 22310

Account number 12421 Account Balance 1000.0 Message done
Receiving transaction 34/71

DEBUG : Server.withdraw - i 35 Current balance 100.0 Amount 500.0 2

DEBUG : Server.processTransactions() - transferring out account 82411

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 82411

Account number 82411 Account Balance -400.0 Message done
Receiving transaction 35/71

DEBUG : Server.withdraw - i 29 Current balance 100.0 Amount 50.0 1

DEBUG : Server.processTransactions() - transferring out account 22310
SERVER 2 received 21117

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 22310

Account number 22310 Account Balance 50.0 Message done
SERVER 1 received 00925

DEBUG : Server.deposit - i 36 Current balance 85.0 Amount 15.0 2
Receiving transaction 36/71

DEBUG : Server.processTransactions() - transferring out account 21117

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21117

Account number 21117 Account Balance 100.0 Message done

DEBUG : Server.deposit - i 37 Current balance 675.0 Amount 325.0 1
Receiving transaction 37/71
SERVER 2 received 31820

DEBUG : Server.withdraw - i 38 Current balance 1900.0 Amount 900.0 2

DEBUG : Server.processTransactions() - transferring out account 31820

DEBUG : Server.processTransactions() - transferring out account 00925
SERVER 2 received 91715

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31820

Account number 31820 Account Balance 1000.0 Message done
Receiving transaction 38/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 00925

Account number 00925 Account Balance 1000.0 Message done
Receiving transaction 39/71

DEBUG : Server.deposit - i 39 Current balance 65.0 Amount 35.0 2

DEBUG : Server.processTransactions() - transferring out account 91715

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91715
SERVER 1 received 32613
SERVER 2 received 13133

Account number 91715 Account Balance 100.0 Message done
Receiving transaction 40/71

DEBUG : Server.deposit - i 41 Current balance 1500.0 Amount 1500.0 1

DEBUG : Server.processTransactions() - transferring out account 32613

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 32613

Account number 32613 Account Balance 3000.0 Message done
Receiving transaction 41/71

DEBUG : Server.withdraw - i 42 Current balance 200.0 Amount 300.0 2

DEBUG : Server.processTransactions() - transferring out account 13133

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 13133
SERVER 1 received 43018

Account number 13133 Account Balance -100.0 Message done
Receiving transaction 42/71
SERVER 2 received 91412

DEBUG : Server.deposit - i 43 Current balance 250.0 Amount 250.0 1

DEBUG : Server.processTransactions() - transferring out account 43018

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 43018

DEBUG : Server.deposit - i 44 Current balance 95.0 Amount 305.0 2

Account number 43018 Account Balance 500.0 Message done
Receiving transaction 43/71
SERVER 1 received 11314

DEBUG : Server.processTransactions() - transferring out account 91412

DEBUG : Server.deposit - i 45 Current balance 0.0 Amount 1000.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91412

DEBUG : Server.processTransactions() - transferring out account 11314

Account number 91412 Account Balance 400.0 Message done
SERVER 2 received 91715
SERVER 1 received 00812
Receiving transaction 44/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 11314

Account number 11314 Account Balance 1000.0 Message done

DEBUG : Server.withdraw - i 39 Current balance 100.0 Amount 150.0 2
Receiving transaction 45/71

DEBUG : Server.processTransactions() - transferring out account 91715

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91715

DEBUG : Server.query - i 46 Current balance 1620.0 1

Account number 91715 Account Balance -50.0 Message done
Receiving transaction 46/71

DEBUG : Server.processTransactions() - transferring out account 00812

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 00812

SERVER 1 received 43817

Account number 00812 Account Balance 1620.0 Message done
Receiving transaction 47/71
SERVER 2 received 61527

DEBUG : Server.deposit - i 48 Current balance 35.0 Amount 65.0 1

DEBUG : Server.processTransactions() - transferring out account 43817

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 43817

DEBUG : Server.withdraw - i 47 Current balance 10000.0 Amount 10000.0 2

Account number 43817 Account Balance 100.0 Message done
Receiving transaction 48/71

DEBUG : Server.processTransactions() - transferring out account 61527

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61527
SERVER 2 received 92716
SERVER 1 received 92018

Account number 61527 Account Balance 0.0 Message done
Receiving transaction 49/71

DEBUG : Server.withdraw - i 51 Current balance 1345.0 Amount 345.0 2

DEBUG : Server.processTransactions() - transferring out account 92716

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92716

DEBUG : Server.withdraw - i 49 Current balance 0.0 Amount 250.0 1

Account number 92716 Account Balance 1000.0 Message done

DEBUG : Server.processTransactions() - transferring out account 92018
Receiving transaction 50/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92018
SERVER 1 received 60716
SERVER 2 received 01113

Account number 92018 Account Balance -250.0 Message done
Receiving transaction 51/71

DEBUG : Server.deposit - i 53 Current balance 300.0 Amount 700.0 1

DEBUG : Server.processTransactions() - transferring out account 60716

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 60716

DEBUG : Server.deposit - i 52 Current balance 145.0 Amount 155.0 2

Account number 60716 Account Balance 1000.0 Message done

DEBUG : Server.processTransactions() - transferring out account 01113

SERVER 1 received 01823

SERVER 2 received 13331

Receiving transaction 52/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 01113

DEBUG : Server.deposit - i 54 Current balance 560.0 Amount 300.0 1

Account number 01113 Account Balance 300.0 Message done

DEBUG : Server.processTransactions() - transferring out account 01823

Receiving transaction 53/71

DEBUG : Server.withdraw - i 55 Current balance 1850.0 Amount 350.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 01823

DEBUG : Server.processTransactions() - transferring out account 13331

SERVER 1 received 92018

SERVER 2 received 81368

Account number 01823 Account Balance 860.0 Message done

Receiving transaction 54/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 13331

Account number 13331 Account Balance 1500.0 Message done

Receiving transaction 55/71

DEBUG : Server.deposit - i 49 Current balance -250.0 Amount 350.0 1

DEBUG : Server.processTransactions() - transferring out account 92018

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92018

DEBUG : Server.withdraw - i 56 Current balance 2350.0 Amount 2000.0 2

Account number 92018 Account Balance 100.0 Message done

Receiving transaction 56/71

DEBUG : Server.processTransactions() - transferring out account 81368

SERVER 1 received 91214

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 81368

SERVER 2 received 91245

Account number 81368 Account Balance 350.0 Message done

DEBUG : Server.deposit - i 57 Current balance 1420.0 Amount 180.0 1

Receiving transaction 57/71

DEBUG : Server.processTransactions() - transferring out account 91214

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91214

Account number 91214 Account Balance 1600.0 Message done

Receiving transaction 58/71

DEBUG : Server.withdraw - i 58 Current balance 3500.0 Amount 2300.0 2

DEBUG : Server.processTransactions() - transferring out account 91245

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91245

Account number 91245 Account Balance 1200.0 Message done

SERVER 2 received 42018

SERVER 1 received 42018

Receiving transaction 59/71

DEBUG : Server.withdraw - i 59 Current balance 980.0 Amount 900.0 2

DEBUG : Server.processTransactions() - transferring out account 42018

DEBUG : Server.withdraw - i 59 Current balance 80.0 Amount 900.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42018

DEBUG : Server.processTransactions() - transferring out account 42018

Account number 42018 Account Balance 80.0 Message done

Receiving transaction 60/71

SERVER 2 received 20922

SERVER 1 received 20922

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42018

Account number 42018 Account Balance -820.0 Message done

Receiving transaction 61/71

DEBUG : Server.withdraw - i 62 Current balance 1740.0 Amount 1500.0 2

DEBUG : Server.processTransactions() - transferring out account 20922

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 20922

DEBUG : Server.withdraw - i 62 Current balance 240.0 Amount 1500.0 1

Account number 20922 Account Balance 240.0 Message done
Receiving transaction 62/71

DEBUG : Server.processTransactions() - transferring out account 20922

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 20922
SERVER 1 received 83830

Account number 20922 Account Balance -1260.0 Message done
SERVER 2 received 83530
Receiving transaction 63/71

DEBUG : Server.withdraw - i 65 Current balance 375.0 Amount 250.0 1

DEBUG : Server.processTransactions() - transferring out account 83830

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 83830

Account number 83830 Account Balance 125.0 Message done
Receiving transaction 64/71

DEBUG : Server.deposit - i 64 Current balance 845250.0 Amount 134750.0 2

DEBUG : Server.processTransactions() - transferring out account 83530
SERVER 1 received 42018

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 83530

Account number 83530 Account Balance 980000.0 Message done
Receiving transaction 65/71

DEBUG : Server.withdraw - i 59 Current balance -820.0 Amount 80.0 1
SERVER 2 received 72820

DEBUG : Server.processTransactions() - transferring out account 42018

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42018

Account number 42018 Account Balance -900.0 Message done
Receiving transaction 66/71

DEBUG : Server.deposit - i 66 Current balance 50000.0 Amount 50000.0 2

DEBUG : Server.processTransactions() - transferring out account 72820
SERVER 1 received 92318
SERVER 2 received 42520

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 72820

Account number 72820 Account Balance 100000.0 Message done
Receiving transaction 67/71

DEBUG : Server.deposit - i 67 Current balance 1290.0 Amount 1110.0 1

DEBUG : Server.processTransactions() - transferring out account 92318

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92318

Account number 92318 Account Balance 2400.0 Message done
Receiving transaction 68/71

DEBUG : Server.deposit - i 68 Current balance 350.0 Amount 650.0 2

DEBUG : Server.processTransactions() - transferring out account 42520
SERVER 1 received 61612
SERVER 2 received 61612

DEBUG : Server.withdraw - i 69 Current balance 625.0 Amount 600.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42520

DEBUG : Server.processTransactions() - transferring out account 61612

Account number 42520 Account Balance 1000.0 Message done
Receiving transaction 69/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61612

Account number 61612 Account Balance 25.0 Message done
Receiving transaction 70/71

Terminating client sending thread - Running time 516 milliseconds

DEBUG : Server.deposit - i 69 Current balance 25.0 Amount 75.0 2

DEBUG : Server.processTransactions() - transferring out account 61612

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61612

Account number 61612 Account Balance 100.0 Message done

Terminating client receiving thread - Running time 618 milliseconds
Server 1 terminating thread - Running time 618 milliseconds
Server 2 terminating thread - Running time 618 milliseconds

Terminating network thread - Client disconnected Server disconnected

Comments:

A synchronized method locks on **this** object meaning each thread will have its own lock, thus not preventing mutual exclusion when multiple threads are active (2 server threads in our case). For static synchronized methods on `Server.class`, you regain mutual exclusion but lose access to instance members like `getServerThreadId()`. Synchronized statements restores both mutual exclusion and access to instance members. Synchronized statement also offer greater control over which part of the code is “locked” ensuring code outside the critical section can be processed concurrently.

Phase 2:

Implementation of Semaphores:

Activating the network ...

Initializing the server ...

Initializing the Accounts database ...

Connecting server to network ...

Initializing client sending application ...

Initializing the transactions ...

Connecting client to network ...

Initializing client receiving application ...

DEBUG : Client.run() - starting client receiving thread connected

Server 1 starting

Server 2 starting

DEBUG : Client.run() - starting client sending thread connected

Receiving transaction 0/71

SERVER 2 received 11528

SERVER 1 received 31718

DEBUG : Server.withdraw - i 1 Current balance 1000.0 Amount 500.0 2

DEBUG : Server.processTransactions() - transferring out account 11528

DEBUG : Server.deposit - i 0 Current balance 10.0 Amount 90.0 1

DEBUG : Server.processTransactions() - transferring out account 31718

SERVER 2 received 16783

SERVER 1 received 10538

DEBUG : Server.withdraw - i 2 Current balance 28000.0 Amount 20000.0 2

DEBUG : Server.processTransactions() - transferring out account 16783

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 11528

SERVER 2 received 21015

DEBUG : Server.withdraw - i 4 Current balance 125.0 Amount 25.0 2

DEBUG : Server.processTransactions() - transferring out account 21015

SERVER 2 received 61326

Account number 11528 Account Balance 500.0 Message done
Receiving transaction 1/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31718

Account number 31718 Account Balance 100.0 Message done
Receiving transaction 2/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 16783

Account number 16783 Account Balance 8000.0 Message done
Receiving transaction 3/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21015

Account number 21015 Account Balance 100.0 Message done
Receiving transaction 4/71

DEBUG : Server.query - i 5 Current balance 1750.0 2

DEBUG : Server.processTransactions() - transferring out account 61326

DEBUG : Server.deposit - i 3 Current balance 120000.0 Amount 20000.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61326
SERVER 2 received 82465

Account number 61326 Account Balance 1750.0 Message done

DEBUG : Server.processTransactions() - transferring out account 10538
Receiving transaction 5/71

DEBUG : Server.deposit - i 6 Current balance 17000.0 Amount 3000.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 10538
SERVER 1 received 31816

Account number 10538 Account Balance 140000.0 Message done

DEBUG : Server.processTransactions() - transferring out account 82465
Receiving transaction 6/71

DEBUG : Server.query - i 7 Current balance 20500.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 82465

DEBUG : Server.processTransactions() - transferring out account 31816
SERVER 2 received 73745

Account number 82465 Account Balance 20000.0 Message done
SERVER 1 received 81514
Receiving transaction 7/71

DEBUG : Server.deposit - i 8 Current balance 75.0 Amount 25.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31816

DEBUG : Server.withdraw - i 9 Current balance 1300.0 Amount 1500.0 1

DEBUG : Server.processTransactions() - transferring out account 73745

DEBUG : Server.processTransactions() - transferring out account 81514

Account number 31816 Account Balance 20500.0 Message done
Receiving transaction 8/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 73745

Account number 73745 Account Balance 100.0 Message done
Receiving transaction 9/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 81514
SERVER 1 received 52853
SERVER 2 received 92012

Account number 81514 Account Balance -200.0 Message done

DEBUG : Server.query - i 12 Current balance 1000.0 1
Receiving transaction 10/71

DEBUG : Server.processTransactions() - transferring out account 52853

DEBUG : Server.deposit - i 11 Current balance 25.0 Amount 725.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 52853

DEBUG : Server.processTransactions() - transferring out account 92012

Account number 52853 Account Balance 1000.0 Message done
SERVER 2 received 41714
Receiving transaction 11/71

DEBUG : Server.deposit - i 13 Current balance 1000.0 Amount 100.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92012

DEBUG : Server.processTransactions() - transferring out account 41714
SERVER 1 received 58342

SERVER 2 received 62080

Account number 92012 Account Balance 750.0 Message done

DEBUG : Server.withdraw - i 14 Current balance 100.0 Amount 100.0 1
Receiving transaction 12/71

DEBUG : Server.processTransactions() - transferring out account 58342

DEBUG : Server.deposit - i 15 Current balance 45.0 Amount 55.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 41714
SERVER 1 received 81514

DEBUG : Server.processTransactions() - transferring out account 62080

Account number 41714 Account Balance 1100.0 Message done
Receiving transaction 13/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 58342

Account number 58342 Account Balance 0.0 Message done
Receiving transaction 14/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 62080

Account number 62080 Account Balance 100.0 Message done
Receiving transaction 15/71

DEBUG : Server.deposit - i 9 Current balance -200.0 Amount 200.0 1

DEBUG : Server.processTransactions() - transferring out account 81514
SERVER 1 received 52726
SERVER 2 received 21638

DEBUG : Server.withdraw - i 17 Current balance 38000.0 Amount 5000.0 1

DEBUG : Server.processTransactions() - transferring out account 52726

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 81514

DEBUG : Server.withdraw - i 16 Current balance 0.0 Amount 1000.0 2

Account number 81514 Account Balance 0.0 Message done

DEBUG : Server.processTransactions() - transferring out account 21638
Receiving transaction 16/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 52726

SERVER 2 received 60520

Account number 52726 Account Balance 33000.0 Message done

SERVER 1 received 91316

Receiving transaction 17/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21638

Account number 21638 Account Balance -1000.0 Message done

Receiving transaction 18/71

DEBUG : Server.deposit - i 19 Current balance 65.0 Amount 1035.0 2

DEBUG : Server.processTransactions() - transferring out account 60520

DEBUG : Server.query - i 18 Current balance 65000.0 1

DEBUG : Server.processTransactions() - transferring out account 91316

SERVER 2 received 41911

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 60520

SERVER 1 received 39331

DEBUG : Server.deposit - i 21 Current balance 1280.0 Amount 220.0 2

DEBUG : Server.processTransactions() - transferring out account 41911

Account number 60520 Account Balance 1100.0 Message done

DEBUG : Server.withdraw - i 22 Current balance 10.0 Amount 10.0 1

DEBUG : Server.processTransactions() - transferring out account 39331

Receiving transaction 19/71

SERVER 2 received 91312

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91316

SERVER 1 received 15307

DEBUG : Server.withdraw - i 23 Current balance 2850.0 Amount 1850.0 2

DEBUG : Server.processTransactions() - transferring out account 91312

Account number 91316 Account Balance 65000.0 Message done

DEBUG : Server.deposit - i 24 Current balance 50.0 Amount 250.0 1

DEBUG : Server.processTransactions() - transferring out account 15307

Receiving transaction 20/71

SERVER 1 received 60520

SERVER 2 received 51728

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 41911

DEBUG : Server.withdraw - i 25 Current balance 0.0 Amount 800.0 2

Account number 41911 Account Balance 1500.0 Message done

DEBUG : Server.processTransactions() - transferring out account 51728

DEBUG : Server.withdraw - i 19 Current balance 1100.0 Amount 100.0 1

DEBUG : Server.processTransactions() - transferring out account 60520

SERVER 2 received 43902

Receiving transaction 21/71

SERVER 1 received 52521

DEBUG : Server.withdraw - i 26 Current balance 1100.0 Amount 100.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 39331

DEBUG : Server.processTransactions() - transferring out account 43902

DEBUG : Server.deposit - i 27 Current balance 5200.0 Amount 1800.0 1

DEBUG : Server.processTransactions() - transferring out account 52521

SERVER 2 received 31215

Account number 39331 Account Balance 0.0 Message done

SERVER 1 received 22310

Receiving transaction 22/71

DEBUG : Server.query - i 28 Current balance 2000.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91312

DEBUG : Server.processTransactions() - transferring out account 31215

Account number 91312 Account Balance 1000.0 Message done

Receiving transaction 23/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 15307

Account number 15307 Account Balance 300.0 Message done

Receiving transaction 24/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 51728

Account number 51728 Account Balance -800.0 Message done

Receiving transaction 25/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 60520

Account number 60520 Account Balance 1000.0 Message done

Receiving transaction 26/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 43902

Account number 43902 Account Balance 1000.0 Message done

Receiving transaction 27/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 52521

Account number 52521 Account Balance 7000.0 Message done

Receiving transaction 28/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31215

Account number 31215 Account Balance 2000.0 Message done

Receiving transaction 29/71

DEBUG : Server.deposit - i 29 Current balance 20.0 Amount 80.0 1

DEBUG : Server.processTransactions() - transferring out account 22310

SERVER 1 received 38421

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 22310

SERVER 2 received 41024

DEBUG : Server.deposit - i 32 Current balance 5300.0 Amount 500.0 1

Account number 22310 Account Balance 100.0 Message done

DEBUG : Server.processTransactions() - transferring out account 38421

DEBUG : Server.deposit - i 31 Current balance 1005.0 Amount 495.0 2

Receiving transaction 30/71

DEBUG : Server.processTransactions() - transferring out account 41024

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 38421

SERVER 1 received 70321

SERVER 2 received 12421

Account number 38421 Account Balance 5800.0 Message done

DEBUG : Server.query - i 33 Current balance 0.0 1

Receiving transaction 31/71

DEBUG : Server.processTransactions() - transferring out account 70321

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 41024

DEBUG : Server.withdraw - i 34 Current balance 1090.0 Amount 90.0 2

Account number 41024 Account Balance 1500.0 Message done

DEBUG : Server.processTransactions() - transferring out account 12421

SERVER 1 received 82411

SERVER 2 received 22310

Receiving transaction 32/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 70321

Account number 70321 Account Balance 0.0 Message done

DEBUG : Server.withdraw - i 35 Current balance 100.0 Amount 500.0 1

Receiving transaction 33/71

DEBUG : Server.processTransactions() - transferring out account 82411

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 12421

DEBUG : Server.withdraw - i 29 Current balance 100.0 Amount 50.0 2

DEBUG : Server.processTransactions() - transferring out account 22310

Account number 12421 Account Balance 1000.0 Message done

SERVER 1 received 00925

SERVER 2 received 21117

Receiving transaction 34/71

DEBUG : Server.deposit - i 37 Current balance 675.0 Amount 325.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 82411

DEBUG : Server.processTransactions() - transferring out account 00925

DEBUG : Server.deposit - i 36 Current balance 85.0 Amount 15.0 2

Account number 82411 Account Balance -400.0 Message done

DEBUG : Server.processTransactions() - transferring out account 21117

SERVER 1 received 31820

SERVER 2 received 91715

Receiving transaction 35/71

DEBUG : Server.withdraw - i 38 Current balance 1900.0 Amount 900.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 22310

DEBUG : Server.processTransactions() - transferring out account 31820

Account number 22310 Account Balance 50.0 Message done
Receiving transaction 36/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 00925

Account number 00925 Account Balance 1000.0 Message done
Receiving transaction 37/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21117

Account number 21117 Account Balance 100.0 Message done
Receiving transaction 38/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 31820

Account number 31820 Account Balance 1000.0 Message done
Receiving transaction 39/71

DEBUG : Server.deposit - i 39 Current balance 65.0 Amount 35.0 2

DEBUG : Server.processTransactions() - transferring out account 91715
SERVER 2 received 13133
SERVER 1 received 32613

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91715

DEBUG : Server.withdraw - i 42 Current balance 200.0 Amount 300.0 2

Account number 91715 Account Balance 100.0 Message done

DEBUG : Server.processTransactions() - transferring out account 13133

DEBUG : Server.deposit - i 41 Current balance 1500.0 Amount 1500.0 1
Receiving transaction 40/71

DEBUG : Server.processTransactions() - transferring out account 32613
SERVER 2 received 43018

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 13133
SERVER 1 received 91412

Account number 13133 Account Balance -100.0 Message done

DEBUG : Server.deposit - i 43 Current balance 250.0 Amount 250.0 2
Receiving transaction 41/71

DEBUG : Server.deposit - i 44 Current balance 95.0 Amount 305.0 1

DEBUG : Server.processTransactions() - transferring out account 43018

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 32613

DEBUG : Server.processTransactions() - transferring out account 91412

Account number 32613 Account Balance 3000.0 Message done
SERVER 1 received 91715
Receiving transaction 42/71
SERVER 2 received 11314

DEBUG : Server.withdraw - i 39 Current balance 100.0 Amount 150.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 43018

DEBUG : Server.processTransactions() - transferring out account 91715

Account number 43018 Account Balance 500.0 Message done

DEBUG : Server.deposit - i 45 Current balance 0.0 Amount 1000.0 2

DEBUG : Server.processTransactions() - transferring out account 11314
Receiving transaction 43/71
SERVER 1 received 00812
SERVER 2 received 61527

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91412

DEBUG : Server.query - i 46 Current balance 1620.0 1

Account number 91412 Account Balance 400.0 Message done

DEBUG : Server.withdraw - i 47 Current balance 10000.0 Amount 10000.0 2
Receiving transaction 44/71

DEBUG : Server.processTransactions() - transferring out account 00812

DEBUG : Server.processTransactions() - transferring out account 61527
SERVER 1 received 43817

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91715
SERVER 2 received 92018

DEBUG : Server.deposit - i 48 Current balance 35.0 Amount 65.0 1

DEBUG : Server.processTransactions() - transferring out account 43817

DEBUG : Server.withdraw - i 49 Current balance 0.0 Amount 250.0 2

Account number 91715 Account Balance -50.0 Message done

DEBUG : Server.processTransactions() - transferring out account 92018

SERVER 1 received 92716

Receiving transaction 45/71

SERVER 2 received 01113

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 11314

DEBUG : Server.withdraw - i 51 Current balance 1345.0 Amount 345.0 1

Account number 11314 Account Balance 1000.0 Message done

DEBUG : Server.processTransactions() - transferring out account 92716

Receiving transaction 46/71

DEBUG : Server.deposit - i 52 Current balance 145.0 Amount 155.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 00812

SERVER 1 received 60716

DEBUG : Server.processTransactions() - transferring out account 01113

Account number 00812 Account Balance 1620.0 Message done

Receiving transaction 47/71

DEBUG : Server.deposit - i 53 Current balance 300.0 Amount 700.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61527

SERVER 2 received 01823

DEBUG : Server.processTransactions() - transferring out account 60716

Account number 61527 Account Balance 0.0 Message done

DEBUG : Server.deposit - i 54 Current balance 560.0 Amount 300.0 2

Receiving transaction 48/71

SERVER 1 received 13331

DEBUG : Server.processTransactions() - transferring out account 01823

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 43817

DEBUG : Server.withdraw - i 55 Current balance 1850.0 Amount 350.0 1

Account number 43817 Account Balance 100.0 Message done
SERVER 2 received 92018

DEBUG : Server.processTransactions() - transferring out account 13331
Receiving transaction 49/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92018

Account number 92018 Account Balance -250.0 Message done
Receiving transaction 50/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92716

Account number 92716 Account Balance 1000.0 Message done
Receiving transaction 51/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 01113

Account number 01113 Account Balance 300.0 Message done
Receiving transaction 52/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 60716

Account number 60716 Account Balance 1000.0 Message done
Receiving transaction 53/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 01823

Account number 01823 Account Balance 860.0 Message done
Receiving transaction 54/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 13331

Account number 13331 Account Balance 1500.0 Message done
Receiving transaction 55/71

DEBUG : Server.deposit - i 49 Current balance -250.0 Amount 350.0 2

DEBUG : Server.processTransactions() - transferring out account 92018
SERVER 1 received 81368

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92018
SERVER 2 received 91214

Account number 92018 Account Balance 100.0 Message done

DEBUG : Server.withdraw - i 56 Current balance 2350.0 Amount 2000.0 1
Receiving transaction 56/71

DEBUG : Server.processTransactions() - transferring out account 81368

DEBUG : Server.deposit - i 57 Current balance 1420.0 Amount 180.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 81368

DEBUG : Server.processTransactions() - transferring out account 91214
SERVER 1 received 91245

Account number 81368 Account Balance 350.0 Message done
Receiving transaction 57/71
SERVER 2 received 42018

DEBUG : Server.withdraw - i 58 Current balance 3500.0 Amount 2300.0 1

DEBUG : Server.processTransactions() - transferring out account 91245

DEBUG : Server.withdraw - i 59 Current balance 980.0 Amount 900.0 2

DEBUG : Server.processTransactions() - transferring out account 42018

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91214
SERVER 1 received 01314
SERVER 2 received 20922

Account number 91214 Account Balance 1600.0 Message done
Receiving transaction 58/71

DEBUG : Server.query - i 61 Current balance 1.05E7 1

DEBUG : Server.processTransactions() - transferring out account 01314

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 91245

DEBUG : Server.withdraw - i 62 Current balance 1740.0 Amount 1500.0 2

Account number 91245 Account Balance 1200.0 Message done

DEBUG : Server.processTransactions() - transferring out account 20922
Receiving transaction 59/71
SERVER 2 received 21219

DEBUG : Server.query - i 63 Current balance 535000.0 2

DEBUG : Server.processTransactions() - transferring out account 21219

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42018
SERVER 2 received 83830

Account number 42018 Account Balance 80.0 Message done
Receiving transaction 60/71
SERVER 1 received 83530

DEBUG : Server.withdraw - i 65 Current balance 375.0 Amount 250.0 2

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 01314

DEBUG : Server.processTransactions() - transferring out account 83830

Account number 01314 Account Balance 1.05E7 Message done
Receiving transaction 61/71

DEBUG : Server.deposit - i 64 Current balance 845250.0 Amount 134750.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 20922
SERVER 2 received 42018

Account number 20922 Account Balance 240.0 Message done

DEBUG : Server.withdraw - i 59 Current balance 80.0 Amount 80.0 2

DEBUG : Server.processTransactions() - transferring out account 83530
Receiving transaction 62/71

DEBUG : Server.processTransactions() - transferring out account 42018

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 21219
SERVER 1 received 72820
SERVER 2 received 92318

DEBUG : Server.deposit - i 66 Current balance 50000.0 Amount 50000.0 1

DEBUG : Server.processTransactions() - transferring out account 72820

DEBUG : Server.deposit - i 67 Current balance 1290.0 Amount 1110.0 2

Account number 21219 Account Balance 535000.0 Message done

DEBUG : Server.processTransactions() - transferring out account 92318
SERVER 1 received 42520
SERVER 2 received 61612
Receiving transaction 63/71

DEBUG : Server.deposit - i 68 Current balance 350.0 Amount 650.0 1

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 83830

DEBUG : Server.withdraw - i 69 Current balance 625.0 Amount 600.0 2

DEBUG : Server.processTransactions() - transferring out account 61612

Account number 83830 Account Balance 125.0 Message done
SERVER 2 received 61612

DEBUG : Server.processTransactions() - transferring out account 42520
Receiving transaction 64/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 83530

Account number 83530 Account Balance 980000.0 Message done
Receiving transaction 65/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42018

Account number 42018 Account Balance 0.0 Message done
Receiving transaction 66/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 72820

Account number 72820 Account Balance 100000.0 Message done
Receiving transaction 67/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 92318

Account number 92318 Account Balance 2400.0 Message done
Receiving transaction 68/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61612

Account number 61612 Account Balance 25.0 Message done
Receiving transaction 69/71

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 42520

Account number 42520 Account Balance 1000.0 Message done
Receiving transaction 70/71

Terminating client sending thread - Running time 523 milliseconds

DEBUG : Server.deposit - i 69 Current balance 25.0 Amount 75.0 2

DEBUG : Server.processTransactions() - transferring out account 61612

DEBUG : Client.receiveTransactions() - receiving updated transaction on account 61612

Account number 61612 Account Balance 100.0 Message done

Terminating client receiving thread - Running time 625 milliseconds

Comments:

The implementation of Semaphores leads to slightly higher run times mainly because it forces a context switch. When Semaphore.acquire() is invoked and the semaphore isn't available, the thread will be blocked and the context switch will occur involving saving the elements of the Program Control Block to be saved to memory and recalled later when the thread is "awoken". An interesting thing about implementing Semaphores instead of busy-waiting is that the buffers are managed by the Semaphores themselves so they serve multiple purposes simplifying the code and ensuring concurrency without deadlocks.